

PERMUTATION & COMBINATION

☆ FACTORIAL NOTATION

$n!$ (n Factorial) : The product of all positive integers from 1 to n .

$$n! = n \times (n-1) \times (n-2) \dots \dots \dots 1$$

Special Case : $0! = 1$

☆ PERMUTATION

Permutation of n objects taken r at a time :

$${}^n P_r = \frac{n!}{(n-r)!}$$

Where :

$n \rightarrow$ Total number of objects

$r \rightarrow$ Number of objects selected

Permutation of n objects where some objects are identical :

$${}^n P_r = \frac{n!}{p_1! \times p_2! \times \dots \times p_k!}$$

Where $p_1, p_2, \dots, p_k$ are the frequencies of identical objects.

☆ COMBINATION

Combination of n objects taken r at a time :

$${}^n C_r = \frac{n!}{r! (n-r)!}$$

Where :

$n \rightarrow$ Total number of objects

$r \rightarrow$ Number of objects selected

RELATION BETWEEN PERMUTATION & COMBINATION

$${}^n P_r = {}^n C_r \times r!$$

PROPERTIES OF PERMUTATION & COMBINATION

☆ SYMMETRIC PROPERTY

$${}^nC_r = {}^nC_{n-r}$$

☆ SUM OF ALL COMBINATION

$$\sum_{r=0}^n {}^nC_r = 2^n$$

☆ ADDITION & MULTIPLICATION RULE

$${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$$

SPECIAL PERMUTATION & COMBINATION

☆ CIRCULAR PERMUTATION

Circular permutation of n objects = $(n-1)!$

☆ Permutation of objects with Repetition = n^r

☆ Combination with Repetition : nC_r (with repetition) = $\frac{(n+r-1)!}{r!(n-1)!}$

BINOMIAL THEOREM

☆ General term of binomial expansion :

$$T_{r+1} = {}^nC_r \cdot a^{n-r} \cdot b^r$$

where : a & b are the terms in the binomial expansion
 nC_r is the binomial coefficient